

2. Performance Benchmark

2a. Isolation Forest-Anomaly Detection

Metric	Value	Description
Precision	99.77%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	92.38%	Share of actual attacks successfully flagged
F1-Score	95.93%	Balanced measure combining Precision and Recall
Accuracy	96.96%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.13%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	7.62%	Actual attacks that went undetected
True Positive Rate (TPR)	92.38%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.87%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 81812	FN 6750
Actual: Normal	FP 185	TN 139254

2b. Random Forest-Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	37829	0	0
Act: ICMP	0	22432	0
Act: UDP	0	0	21546

2c. Response Latency

Metric	Value	Description
Detection Time	501.9 ms	Average time taken to detect an attack
Mitigation Response Time	0.7 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.07%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	2.66%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	2.62%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

No offence history for this period.

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 14:29:23	10.0.0.3	DDoS	Uncertain	59.7%	Low	Released
2026-09-04 14:29:23	10.0.0.5	DDoS	Uncertain	59.7%	Low	Released
2026-09-04 14:31:54	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:31:55	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:32:00	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:32:00	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-04 14:32:00	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:32:01	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-04 14:32:01	10.0.0.17	DDoS	SYN Flood	90.1%	Critical	Time Ban (1m)
2026-09-04 14:32:01	10.0.0.17	DDoS	SYN Flood	90.1%	Critical	Time Ban
2026-09-04 14:32:05	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban (1m)
2026-09-04 14:32:06	10.0.0.18	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-04 14:32:06	10.0.0.19	DDoS	SYN Flood	92.8%	Critical	Time Ban (1m)
2026-09-04 14:32:06	10.0.0.19	DDoS	SYN Flood	92.8%	Critical	Time Ban
2026-09-04 14:33:00	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (2m)
2026-09-04 14:33:01	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (2m)
2026-09-04 14:33:05	10.0.0.18	DDoS	SYN Flood	91.1%	High	Time Ban (2m)
2026-09-04 14:33:06	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-04 14:33:12	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-04 14:35:00	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (5m)
2026-09-04 14:35:01	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (5m)
2026-09-04 14:35:05	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (5m)
2026-09-04 14:35:12	10.0.0.19	DDoS	SYN Flood	94.0%	Critical	Time Ban (5m)
2026-09-04 14:39:28	10.0.0.1	DDoS	Uncertain	59.7%	Low	Released
2026-09-04 14:39:28	10.0.0.4	DDoS	Uncertain	58.8%	Low	Released
2026-09-04 14:41:54	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:41:55	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:41:55	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:42:15	10.0.0.24	DDoS	ICMP Flood	91.8%	High	Time Ban (1m)
2026-09-04 14:42:15	10.0.0.24	DDoS	ICMP Flood	91.8%	Critical	Time Ban
2026-09-04 14:42:16	10.0.0.25	DDoS	ICMP Flood	91.8%	High	Time Ban (1m)
2026-09-04 14:42:16	10.0.0.23	DDoS	ICMP Flood	91.9%	High	Time Ban (1m)
2026-09-04 14:42:16	10.0.0.23	DDoS	ICMP Flood	92.0%	Critical	Time Ban
2026-09-04 14:42:16	10.0.0.25	DDoS	ICMP Flood	91.9%	High	Time Ban
2026-09-04 14:43:15	10.0.0.24	DDoS	ICMP Flood	92.2%	High	Time Ban (2m)
2026-09-04 14:43:16	10.0.0.25	DDoS	ICMP Flood	92.6%	High	Released
2026-09-04 14:43:16	10.0.0.23	DDoS	ICMP Flood	92.6%	High	Released

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-04 14:43:22	10.0.0.25	DDoS	ICMP Flood	92.6%	Critical	Time Ban (2m)
2026-09-04 14:43:22	10.0.0.23	DDoS	ICMP Flood	92.6%	Critical	Time Ban (2m)
2026-09-04 14:45:15	10.0.0.24	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 14:45:28	10.0.0.25	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 14:45:28	10.0.0.23	DDoS	ICMP Flood	96.0%	Critical	Time Ban (5m)
2026-09-04 14:51:54	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:51:55	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:51:56	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-04 14:52:04	10.0.0.22	DDoS	UDP Flood	79.1%	Critical	Time Ban (1m)
2026-09-04 14:52:04	10.0.0.22	DDoS	UDP Flood	79.1%	High	Time Ban
2026-09-04 14:52:15	10.0.0.20	DDoS	UDP Flood	76.0%	High	Time Ban (1m)
2026-09-04 14:52:15	10.0.0.20	DDoS	UDP Flood	76.2%	High	Time Ban
2026-09-04 14:52:16	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-04 14:52:16	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-04 14:53:04	10.0.0.22	DDoS	UDP Flood	83.6%	Critical	Time Ban (2m)
2026-09-04 14:53:15	10.0.0.20	DDoS	UDP Flood	78.9%	Critical	Time Ban (2m)
2026-09-04 14:53:16	10.0.0.21	DDoS	UDP Flood	77.9%	Critical	Time Ban (2m)
2026-09-04 14:55:04	10.0.0.22	DDoS	UDP Flood	83.6%	Critical	Time Ban (5m)
2026-09-04 14:55:15	10.0.0.20	DDoS	UDP Flood	78.9%	Critical	Time Ban (5m)
2026-09-04 14:55:16	10.0.0.21	DDoS	UDP Flood	77.9%	High	Time Ban (5m)
2026-09-04 15:02:15	10.0.0.25	DDoS	ICMP Flood	89.4%	Critical	Blackhole
2026-09-04 15:02:36	10.0.0.23	DDoS	ICMP Flood	90.3%	Critical	Blackhole
2026-09-04 15:03:02	10.0.0.19	DDoS	SYN Flood	92.5%	Critical	Blackhole
2026-09-04 15:05:00	10.0.0.16	DDoS	SYN Flood	92.7%	Critical	Blackhole
2026-09-04 15:05:03	10.0.0.18	DDoS	SYN Flood	92.9%	Critical	Blackhole
2026-09-04 15:05:36	10.0.0.24	DDoS	ICMP Flood	96.0%	Critical	Blackhole
2026-09-04 15:13:59	10.0.0.17	DDoS	SYN Flood	93.4%	Critical	Blackhole
2026-09-04 15:14:25	10.0.0.21	DDoS	Uncertain	-	Low	Released
2026-09-04 15:14:33	10.0.0.21	DDoS	UDP Flood	75.9%	Critical	Time Ban (10m)
2026-09-04 15:14:34	10.0.0.20	DDoS	UDP Flood	74.2%	Critical	Time Ban (10m)
2026-09-04 15:15:08	10.0.0.22	DDoS	UDP Flood	79.6%	Critical	Blackhole

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE